GDS2 WAFER MAP CONVERTER

User Manual & Technical Reference — Version 1.0

Semiconductor Wafer Map Generation

GDS2 ASCII Text → SINF / KLA Compatible Wafer Map

Python 3.8+

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1. Overview

The GDS2 Wafer Map Converter is a desktop application that transforms GDS2-format layout files into ASCII wafer maps compatible with SINF / KLA-format test systems used in semiconductor manufacturing.

The tool provides a visual preview of the wafer die map, allows interactive cell editing, supports custom bin definitions, and exports production-ready wafer map text files.

1.1 Key Capabilities

- Parse GDS2 ASCII text files and extract die placement coordinates
- Auto-detect die pitch (X and Y) from coordinate data
- Render an interactive colour-coded wafer map
- Allow per-cell editing by clicking directly on the map canvas
- Edit the raw output text and map in real time
- Open existing ASCII wafer maps for review and re-export

1.2 System Requirements

Requirement	Detail
Operating System	Windows 10 / 11, macOS 12+, or Linux (Ubuntu 20.04+)
Python Version	Python 3.8 or later

2. Step 1 — Preparing the GDS2 ASCII Text File

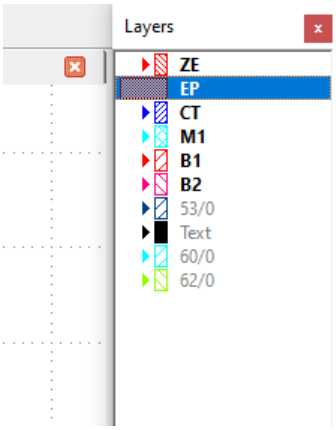
Before the GDS2 Wafer Map Converter can process your layout, the binary GDS file (.GDS /.GDS2) must first be exported as a GDS2 text file. This is a mandatory pre-processing step and is performed in your EDA / layout tool (KLayout), not within this application.

Why is this step needed?

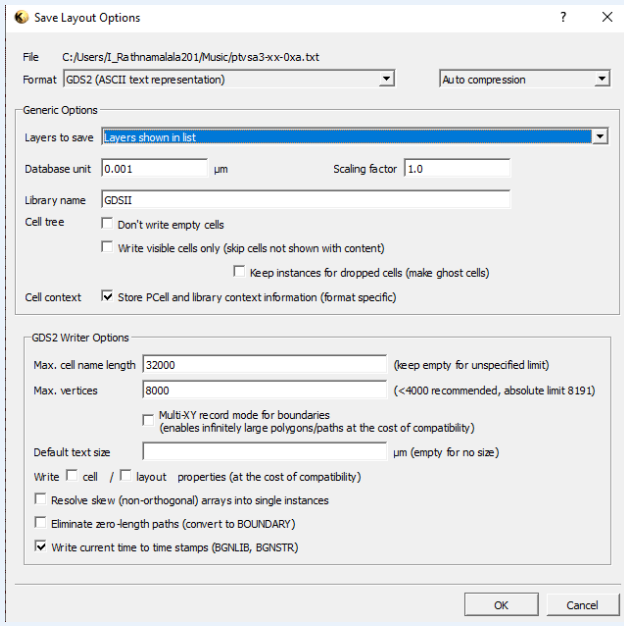
GDS2 files are stored in a compact binary format that the converter cannot read directly.
The text export 'unpacks' the binary records into plain text with one record per line,
which the converter then parses to locate SREF (structure reference) die coordinates.

2.1 Exporting from KLayout

1. **Open** your GDS file in KLayout.
2. **Select the layer** that needs to be exported. (To minimize export file size)



3. Navigate **to File → Save As → Save as type → Select GDS2 Text files → Save**
4. Under **Layers to save** select **layers shown in list**.
5. **Click OK** to run the export. The resulting .txt file is your input for this tool.



KLayout —ASCII Export dialog

2.2 Verifying the ASCII File

Before opening the file in the converter, open it in a text editor and confirm:

- The first line begins with HEADER
- There are SREF records followed by SNAME and XY lines
- XY values appear as integers (in database units, typically nanometres)

Example of a valid ASCII GDS2 snippet:

```
HEADER 600
BGNLIB ...
LIBNAME my design
SREF
  SNAME z5_subdef1
  XY 1473000 : 1473000
ENDEL
```

3. Launching the Application

3.1 Starting the Application

1. Run: ***GDS2 Wafer Map Converter.exe***

3.2 Application Layout

The window is divided into three primary areas:

- Left Sidebar — all settings, parameters, actions, statistics and bin definitions
- Centre — interactive map canvas with zoom, pan and cell-click editing
- Right Panel — editable raw output text that updates the map in real time

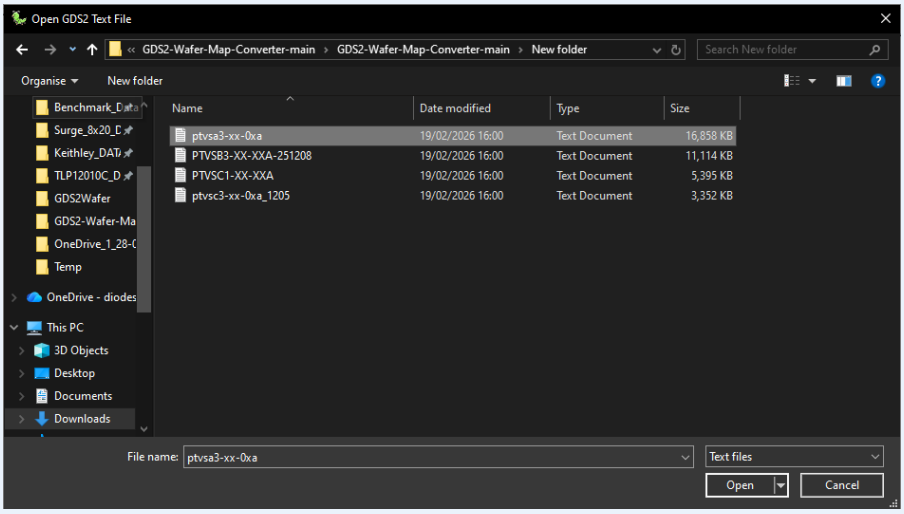
4. Step 2 — Loading the GDS2 ASCII File

4.1 Opening the File

Use one of the following methods to load the ASCII GDS2 text file prepared in Step 1:

- **Toolbar: click the 🗁 Open button**
- **Menu: File → Open GDS2 File**
- **Keyboard shortcut: Ctrl + O**

A standard file browser dialog will open. Navigate to your .txt GDS2 ASCII file and click Open.



File open dialog — selecting a GDS2 ASCII .txt file

4.2 After Loading

Once loaded, the application will:

- Display the filename in the Input File box in the sidebar
- Enable the Convert button and toolbar icon
- Show the file size in the status bar

The file is held in memory. No conversion takes place until you click Convert.

5. Step 3 — Configuring Wafer Settings

5.1 Wafer Settings Reference

Field	Description
Wafer ID	Identifier string written into the output file header. Defaults to GDS2_WAFER_MAP.
Diameter (mm)	Physical wafer diameter in millimetres. Common values: 100, 150, 200, 300.
Die Size X (mm)	Die width in millimetres, used to compute the die grid column pitch.
Die Size Y (mm)	Die height in millimetres, used to compute the die grid row pitch.
Die Structures	Comma-separated list of SNAME values to include. Only SREFs whose SNAME matches will be counted as dies.
Show Edge Dies (*)	When checked, wafer positions adjacent to active dies but outside the circle are marked as edge / scribe cells.

5.2 Auto Detecting Die Pitch

If you do not know the exact die size, ***click Auto-detect***. The tool will analyse the X and Y coordinate intervals in the loaded GDS2 file and populate the Die Size X and Die Size Y fields automatically.

> **Tip**
> Auto-detect works best when the GDS2 file contains a regular array of SREF placements.
> For irregular arrays, manually verify the detected values against your process specification.

5.3 Identifying Die Structure Names

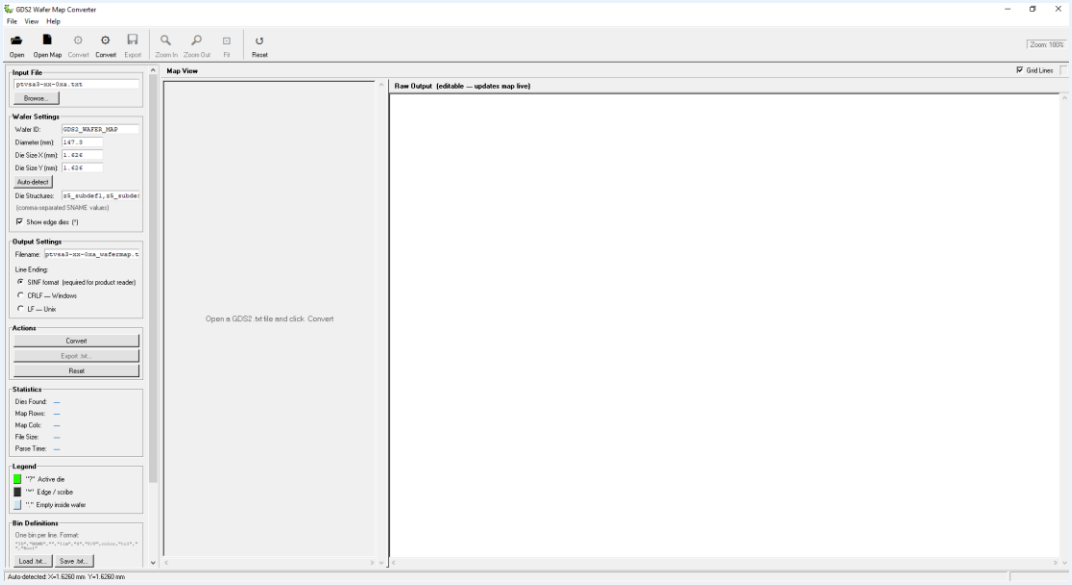
The Die Structures field is critical. Open your GDS2 ASCII file in a text editor and search for SNAME lines to find the exact names used for die-level cell placements. Only those names listed here will be recognised as dies.

Example: if your GDS2 file contains SNAME z5_subdef1 and SNAME z5_subdef2, enter: z5_subdef1, z5_subdef2

6. Step 4 — Running the Conversion

6.1 Starting Conversion

Click the Convert button in the Actions section of the sidebar or use the ⚙ Convert toolbar button.



Conversion in progress — status bar message and disabled buttons

6.2 Conversion Process

The converter performs the following steps internally:

1. Parses all SREF records from the GDS2 ASCII text, filtering by the Die Structures names.
2. Converts raw XY coordinates from database units (nanometres) to millimetres by dividing by 1,000,000.
3. Aligns each coordinate to the nearest die grid position using the specified X and Y pitch.
4. Applies the wafer boundary circle (diameter) to exclude dies outside the physical wafer.
5. Builds the 2D ASCII grid of '?', '*' and '.' cells.
6. Formats the output file with the SINF-compatible header and optional bin table rows.

6.3 Statistics Panel

After a successful conversion the Statistics panel updates with:

Statistic	Meaning
Dies Found	Total number of die coordinates extracted from the GDS2 file.
Map Rows	Number of rows in the generated grid.
Map Cols	Number of columns in the generated grid.
File Size	Size of the generated output in bytes / KB.
Parse Time	Total elapsed time for the conversion in seconds.

6.4 Troubleshooting Failed Conversion

No die coordinates found

The Die Structures field does not match any SNAME in the file.
Open the GDS2 ASCII file in a text editor, search for 'SNAME', and copy the exact values.

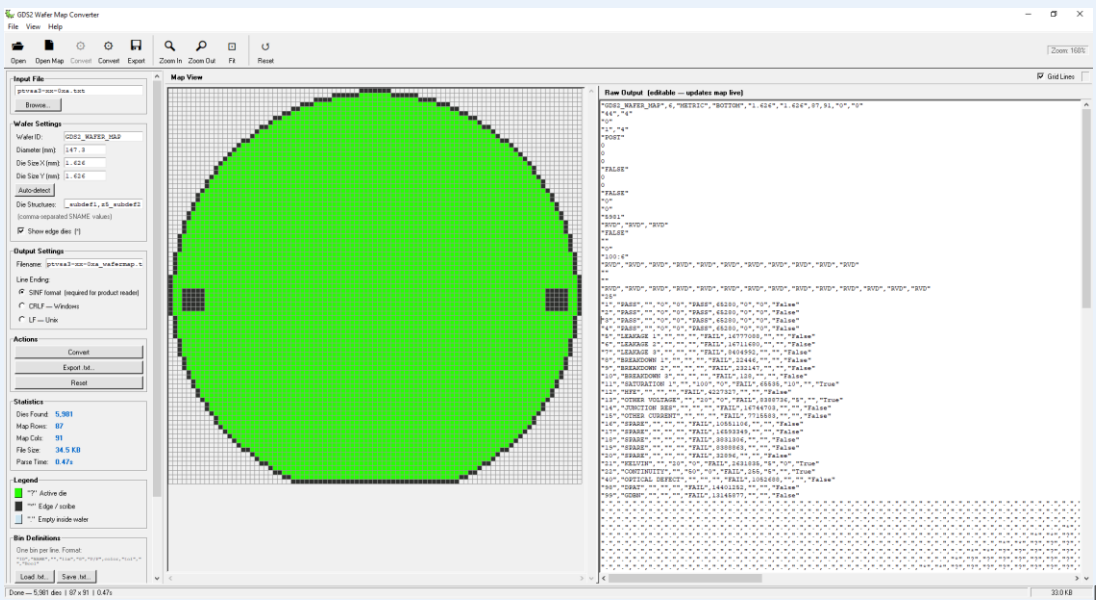
Invalid setting error

A non-numeric value has been entered in Diameter, Die Size X or Die Size Y.
Ensure these fields contain decimal numbers only (e.g. 147.3, not '147.3mm').

7. Using the Map View

7.1 Colour Legend

Symbol / Colour	Meaning
“?” (Green)	Active die — a valid die placement found within the wafer boundary.
“*” (Grey)	Edge / scribe die — adjacent to active dies, at or near the wafer edge.
“.” (Empty)	Empty — inside the wafer boundary but no die placed here.



Map view showing full wafer — green dies, edge cells, empty positions

7.2 Navigation

Action	Effect
Scroll wheel	Pan the map vertically.
Ctrl + Scroll wheel	Zoom in / zoom out.
Toolbar Zoom In / Out	Step zoom using buttons.
Toolbar Fit	Fit the entire wafer map into the visible canvas area.
Grid Lines toggle	Show or hide the cell boundary grid lines (clearer at high zoom).
Hover over cell	Displays the column, row and cell type in the top-right coordinate label.

7.3 Editing Individual Cells

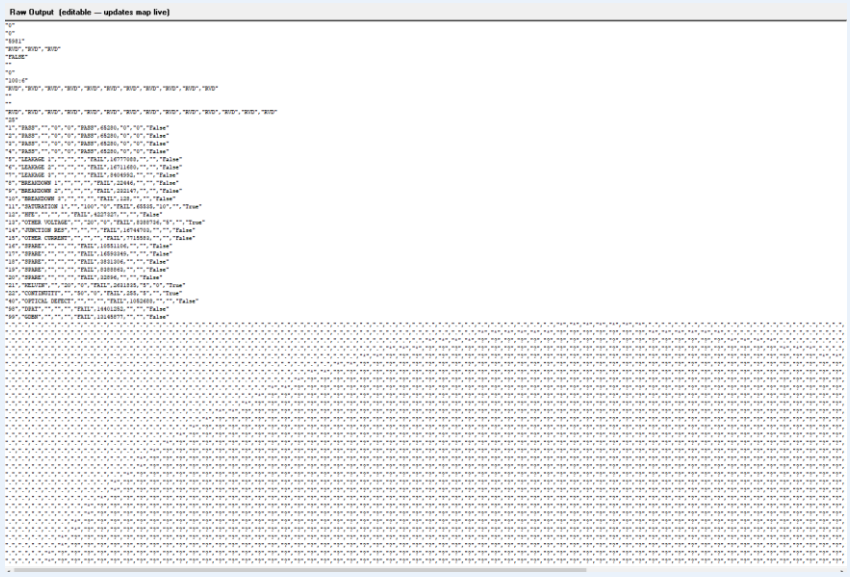
Click any cell on the map to cycle its value through the three states:

Active Die (?) → Edge (*) → Empty (.) → Active Die (?)

Each click updates both the canvas and the Raw Output panel simultaneously so the exported file reflects the change.

8. Raw Output Panel

The right-hand panel shows the full text content of the wafer map file that will be exported. It is directly editable.



Raw Output panel showing SINF-format header and map rows

8.1 Live Editing

Any change typed into the Raw Output panel is reflected on the map canvas.

This allows you to:

- Manually change individual cell values by editing the quoted characters
- Modify header fields (Wafer ID, row/col counts) directly in the file
- Paste in a map from another source and have it rendered immediately

8.2 Text Font Zoom

Hold Ctrl and scroll the mouse wheel inside the Raw Output panel to increase or decrease the font size for easier reading of dense content.

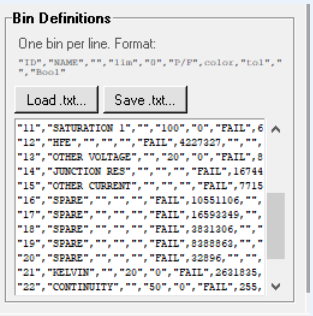
8.3 Opening an Existing Wafer Map

To load and view a previously exported or third-party ASCII wafer map file without performing a GDS2 conversion:

- Use File → Open ASCII Wafer Map... or the Open Map toolbar button
- The file content is loaded into the Raw Output panel and the map is rendered automatically
- The Convert button is disabled in this mode; Export remains available

9. Bin Definitions

The Bin Definitions section at the bottom of the sidebar allows you to define the pass/fail bin table that is embedded in the exported wafer map file. A default set of 25 bins is pre-loaded.



Bin Definitions editor with default bin table visible

9.1 Bin Format

Each line defines one bin in the following format:

Bin Line Format:

"ID","NAME","","limit","0","PASS/FAIL",color_integer,"tolerance","","Bool"

Example: "1","PASS","","0","0","PASS",65280,"0","0","False"

Example: "5","LEAKAGE 1","","","","FAIL",16777088,"","","False"

9.2 Loading and Saving Bin Templates

- **Click Load .txt** to **import** a previously saved bin table from a text file
- **Click Save .txt** to **export** the current bin table for reuse in future sessions

10. Step 5 — Exporting the Wafer Map

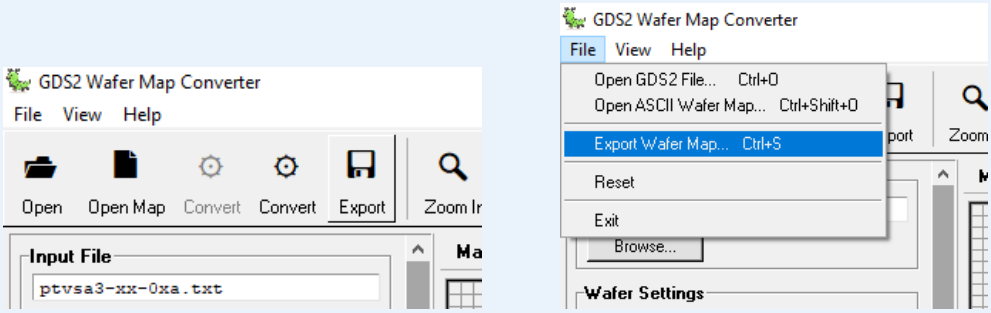
10.1 Output Settings

Setting	Description
Filename	Default output filename, pre-filled from the input file name. Change as required.
SINF format	Recommended. Each map row ends with ,"\r" (hex 22 0D 22 0D 0A) as required by SINF-compatible readers.
CRLF — Windows	Standard Windows line endings (0D 0A). Use if your reader requires plain CRLF.
LF — Unix	Unix line endings (0A only). Use for Linux-based analysis tools.

10.2 Saving the File

Click Export .txt in the Actions section or use **File → Export Wafer Map** (Ctrl+S). A save dialog will appear. Choose a destination folder and filename, then click Save.

A confirmation dialog will show the full path and file size on success.



Export methods

11. Complete Workflow Summary

Step	Action
1	Export the GDS binary file to GDS2 text format using your EDA tool (KLayout, IC Compiler, or Virtuoso).
2	Launch the application: GDS2 Wafer map Converter
3	Click Open (Ctrl+O) and select the GDS2 ASCII .txt file.
4	Set Wafer ID, Diameter, Die Size X/Y, and Die Structures. Use Auto-detect if unsure of pitch.
5	Click Convert. Review the map and statistics.
6	Optionally edit cells on the map canvas or in the Raw Output panel.
7	Review and customise the Bin Definitions table if required.
8	Select the line ending format and click Export .txt...

12. Keyboard Shortcuts & Tips

Shortcut	Action
Ctrl + O	Open GDS2 file
Ctrl + Shift + O	Open ASCII wafer map file
Ctrl + S	Export wafer map
Ctrl + Scroll (canvas)	Zoom in / out on the map
Ctrl + Scroll (raw text)	Increase / decrease text font size
Click on map cell	Cycle cell value: Die → Edge → Empty → Die

13. Frequently Asked Questions

- **The map shows no dies after conversion.**
  - *The SNAME values in the Die Structures field do not match those in the GDS2 file. Open the ASCII file in a text editor, search for 'SNAME', and copy the exact values into the field.*

- **Auto-detect returns unexpected pitch values.**
  - *Ensure the Die Structures field is populated before clicking Auto-detect. The pitch detection analyses only the coordinates of the specified structure names.*

- **The wafer circle looks too large or small.**
  - *Verify the Diameter value is correct for your wafer size. Typical values: 100 mm (4"), 150 mm (6"), 200 mm (8"), 300 mm (12").*

- **Export produces an empty or very small file.**
  - *Ensure conversion has been run and the Statistics panel shows a non-zero die count before exporting.*

- **Can I use the tool without a GDS2 file?**
  - *Yes. Use File → Open ASCII Wafer Map to load an existing wafer map text file directly. The map will render and can be edited and re-exported.*

14. Glossary

Term	Definition
GDS2	Graphic Database System II — industry-standard IC layout file format.
SINF	Standard Interface Format — wafer map format used by KLA and other test equipment.
SREF	Structure Reference — a GDS2 record type that places one cell at a specified coordinate.
SNAME	Structure Name — the name of the cell being referenced by an SREF record.
Die Pitch	The centre-to-centre distance between adjacent dies, equal to die width (X) and die height (Y).
Edge Die (*)	A wafer position adjacent to an active die but at or near the wafer boundary.
Database Units	The unit system used inside a GDS2 file, typically nanometres (1 µm = 1,000 units).
KLA	KLA Corporation — manufacturer of wafer inspection and metrology equipment.